A1 production kernel manifest – verifier reads stored fields + calls backend D(k) (v1.2)

TECT verification-first repository · claim A1-PRODUCTION-KERNEL-MANIFEST

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1. Purpose and scope

A production config may be cited as the numerical implementation of A2/A3 only if its stored kernel parameters are mutually consistent AND it lies in the analytic branch. v1.2 makes the verifier actually read the stored fields and call the backend kernel, closing the two v1.1 implementation gaps (hardcoded q_0, derived Y, no backend evaluation). Verification: `codes/foundations/a1_kernel_checks.py` v1.2 (9/9 PASS); config `claims/A1-PRODUCTION-KERNEL-MANIFEST/canonical_n001_kernel.json`.

2. The schema (unchanged from v1.1) and the verifier (new in v1.2)

A config stores five independent physical fields `r_zero` $:= D(0)$, `mu2_shell` $:= D(q_\star)$, `q0` $:= q_\star$, Z, Y, plus separate tolerances $\varepsilon_Z, \varepsilon_m, \varepsilon_r$ and backend provenance. Gates:

$$\delta_Z := \left|Z + 2Y\,\mathtt{q0}^2\right|, \quad \delta_m := \left|\mathtt{mu2\_shell} - \left(\mathtt{r\_zero} - \tfrac{Z^2}{4Y}\right)\right|, \quad \delta_r := \left|\mathtt{r\_zero} - \left(\mathtt{mu2\_shell} + Y\,\mathtt{q0}^4\right)\right|. \tag{1}$$

$$\mathtt{A1\text{-}KERNEL\text{-}CONSISTENCY\text{-}MANIFEST}: \quad Y > 0,\ Z < 0,\ \delta_Z \le \varepsilon_Z,\ \delta_m \le \varepsilon_m,\ \delta_r \le \varepsilon_r, \tag{2}$$

$$\mathtt{A1\text{-}ANALYTIC\text{-}BRANCH\text{-}MANIFEST}: \quad \mathtt{A1\text{-}KERNEL\text{-}CONSISTENCY\text{-}MANIFEST}\ +\ \mathtt{mu2\_shell} > 0. \tag{3}$$

The legacy alias `mu2` $= r$ is FORBIDDEN; `mu2_shell` is required. The verifier reads all five fields AS-IS from the config JSON (no reconstruction, no hardcoding) and computes $\delta_Z, \delta_m, \delta_r$ from them.

3. Semantic backend check (v1.2)

The verifier imports the backend module (`Math424_AddA_reading_uniqueness`) and evaluates the backend kernel

$$D_{\mathrm{code}}(k) = \mathtt{MU2\_CANON} + C\,(k^2 - Q_0^2)^2 \qquad (\mathtt{MU2\_CANON}, C, Q_0 \text{ imported constants}), \tag{4}$$

verifying $D_{\mathrm{code}}(k) = \mathtt{r\_zero} + Z|k|^2 + Y|k|^4$ at $k \in \{0, 0.31, q_0, 0.93, 1.4, 2.2\}$ to 4.4×10^{-16} (assert `backend_Dcode_matches_stored_params`). A source hash alone is only provenance; this is an implementation-semantics check. That the reader has teeth is demonstrated: corrupting the stored q_0 by 1% makes the residual $9.0 \times 10^{-2} \gg$ tol (assert `stored_field_error_is_detected`). The pinned `backend_sha256` is retained as provenance.

4. Canonical N-001 candidate and the legacy template

The canonical N-001 config (`r_zero` $= 0.21903$, `mu2_shell` $= 0.005$, `q0` $= 0.6801748$, $Z = -0.92528$, $Y = 1$) passes both manifests ($\delta_Z = \delta_r = 0$, $\delta_m = 4.3 \times 10^{-18}$, `mu2_shell` > 0): the NUMERICAL GATES PASS but OPERATOR CERTIFICATION IS PENDING — the card stays T1 OPEN and the config is NOT yet citable as the A2/A3 implementation. The illustrative template $(r, Z, Y, q_0) = (0.35, -1, 0.50, 0.6801748)$ FAILS ($\delta_Z = 0.537$, derived `mu2_shell` $= -0.15 < 0$, `mu2` $= r$ alias).

5. Devil's-advocate

(α) *"Gates pass, so close it."* VALID direction, OPEN retained: closure is an operator act (ratify $\varepsilon_Z, \varepsilon_m, \varepsilon_r$ + the config/backend hashes). The script now reports "certification pending," not "citable." (β) *"Hardcoded q_0 again?"* DISMISSED — v1.2 reads q_0 and Y from the JSON and the corrupted-q_0 test fires, proving the read is load-bearing. (γ) *"Equal tolerances merged?"* DISMISSED — $\varepsilon_Z, \varepsilon_m, \varepsilon_r$ are separate JSON fields; they are intentionally equal (10^{-9}, same dimensionless TECT energy-scale class) and documented as such in the config `tolerance_note`. Quantitative sanity check: semantic residual 4.4×10^{-16}; corrupted-q_0 residual 9.0×10^{-2} (14 orders of magnitude separation).

6. Result footer

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Result ID:               A1-PRODUCTION-KERNEL-MANIFEST (OPEN, v1.2)
Precise statement:       Verifier reads 5 stored fields (r_zero, mu2_shell, q0,
                         Z, Y) AS-IS from canonical_n001_kernel.json; computes
                         delta_Z,delta_m,delta_r (separate eps); imports the
                         backend module and verifies D_code(k)=r_zero+Z k^2+Y k^4
                         (semantic) + corrupted-q0 detection + pinned sha256.
                         CONSISTENCY: Y>0,Z<0,deltas. ANALYTIC-BRANCH: +mu2_shell
                         >0 (only A2/A3-citable). Alias mu2=r FORBIDDEN.
Scope:                   production-config certification; two nested manifests.
Dependencies:            A1-KERNEL-IDENTITY; A1-SCALAR-ANALYTIC-BRANCH.
Evidence grade:          ANALYTIC (gates) + EXECUTED (9/9, semantic + provenance).
Reproduction command:    python codes/foundations/a1_kernel_checks.py
Expected output:         9/9 PASS; canonical gates pass (cert pending); template
                         FAILS; corrupted-q0 detected.
Tier:                    T1 OPEN (numerical gates pass; certification pending).
Falsifier:               operator-certified N-001 manifest (ratified tol + hashes)
                         closes OPEN; any config failing a delta is rejected.
No-overclaim:            gates passing != certification; canonical config NOT yet
                         operator-certified, NOT yet citable as A2/A3 impl;
                         condensate-branch configs do NOT support the A2/A3 gap.
Next required action:    Operator certifies the N-001 ANALYTIC-BRANCH manifest
                         (ratify eps_Z,eps_m,eps_r + config/backend sha256).
```